

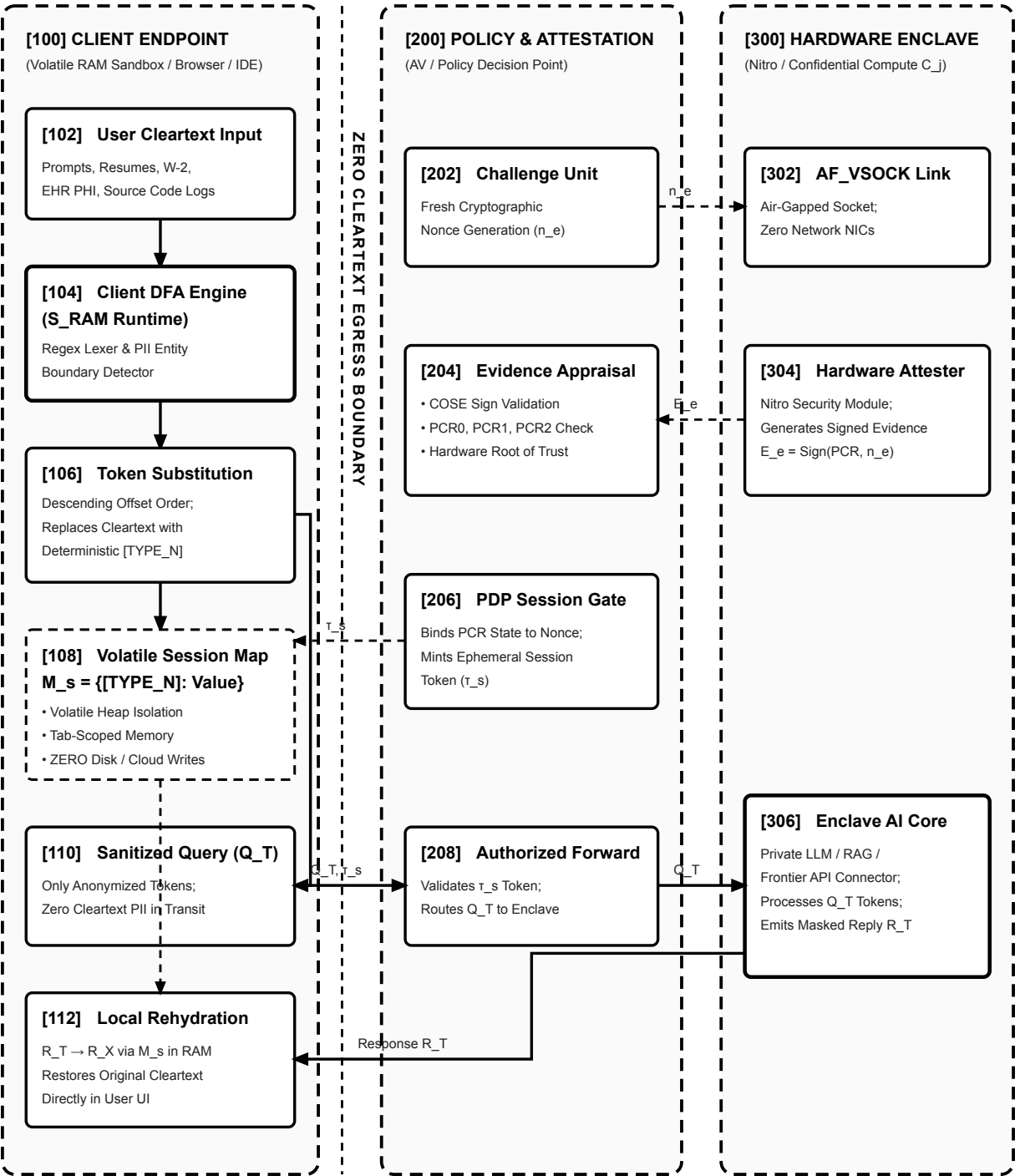


FIG. 1
High-Level System Architecture Diagram Illustrating Three Decoupled Trust Planes: Client Endpoint (100), Attestation/Policy Plane (200), and Hardware Confidential Enclave (300).

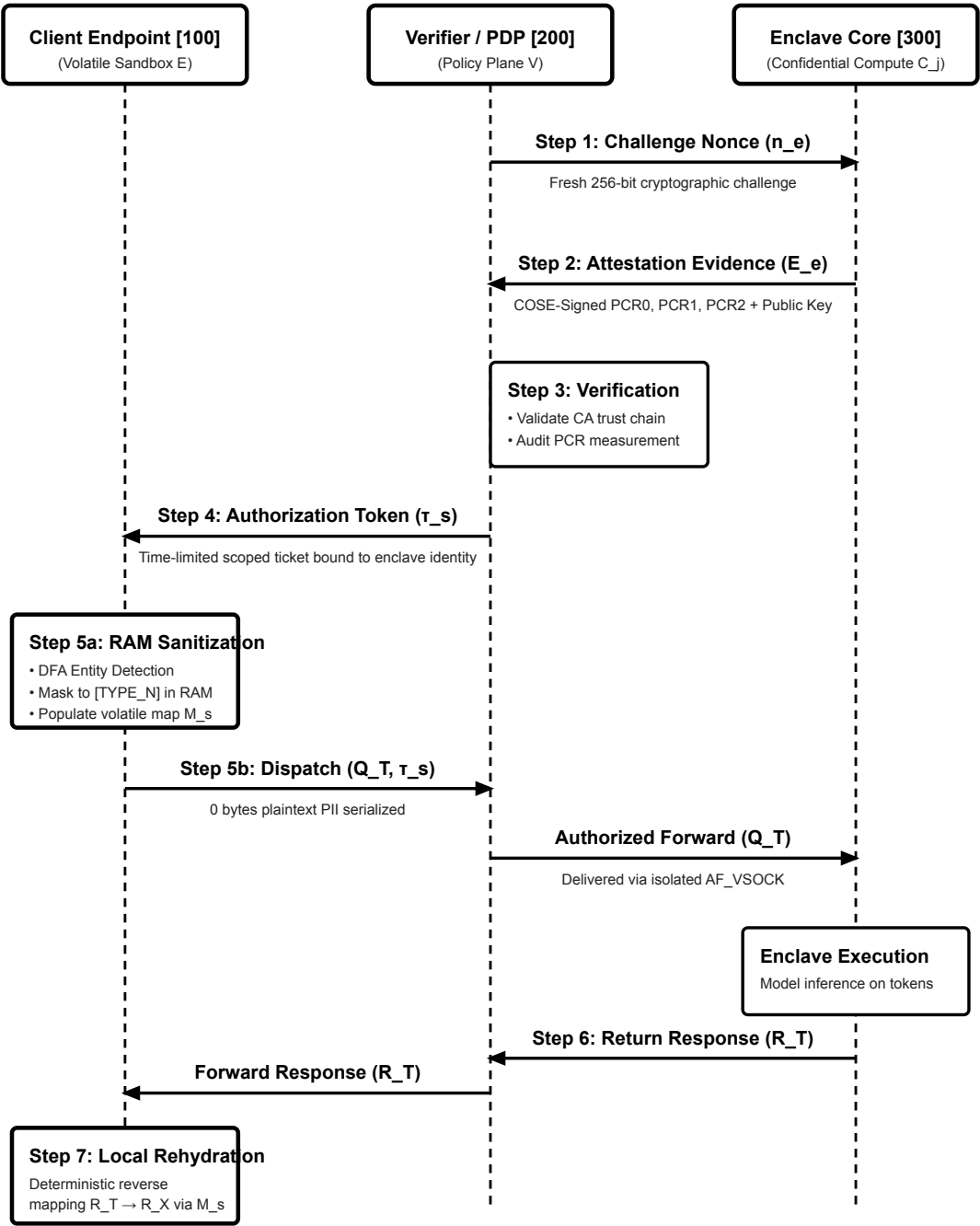


FIG. 2
Sequence Flow Diagram Illustrating the 7-Phase Remote Hardware Attestation and Fail-Closed Workload Authorization Protocol.

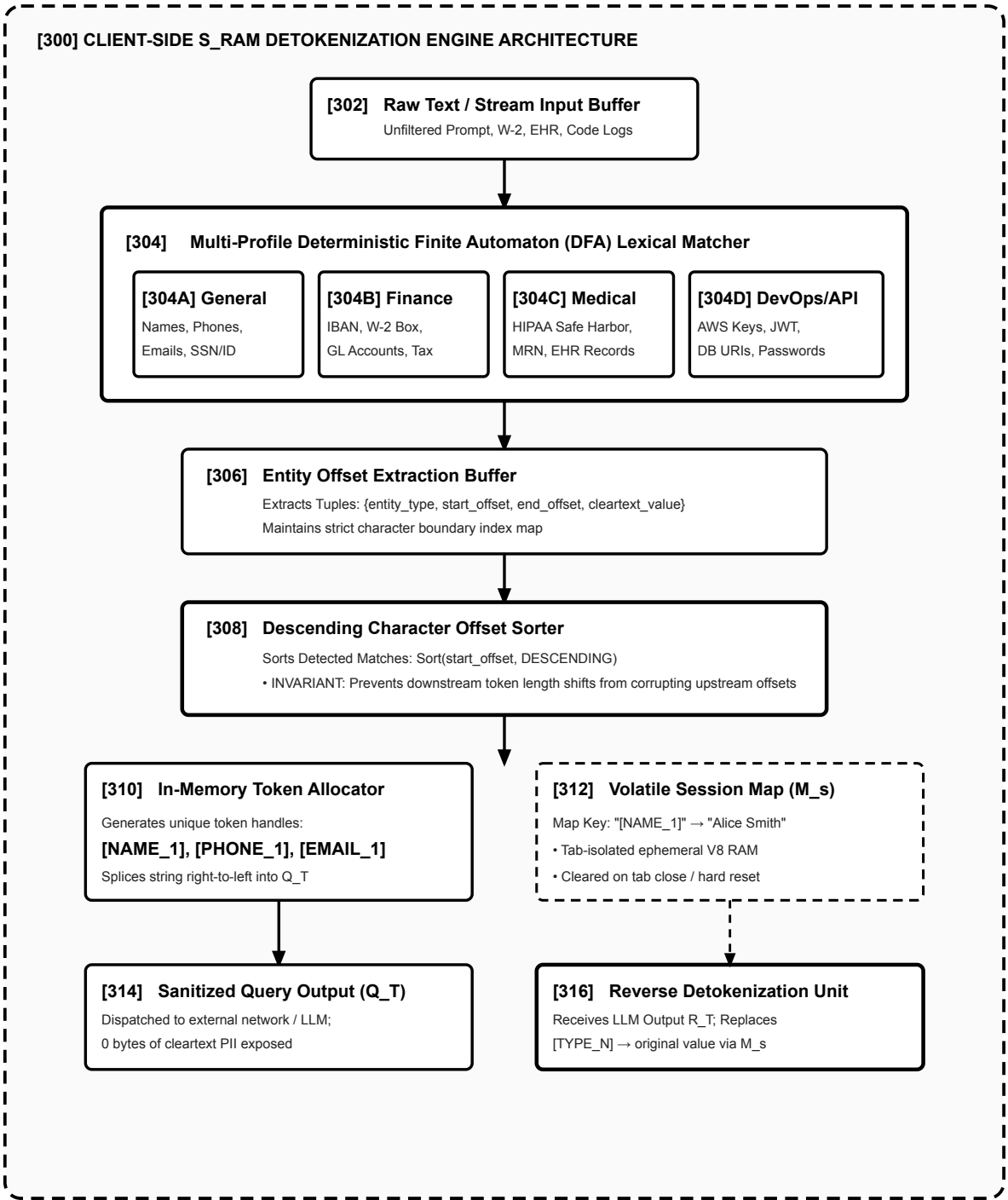


FIG. 3

Schematic Block Diagram Illustrating the Client-Side DFA Tokenization Engine, Descending Character Offset Sorter (308), and Volatile Heap Isolation (312).

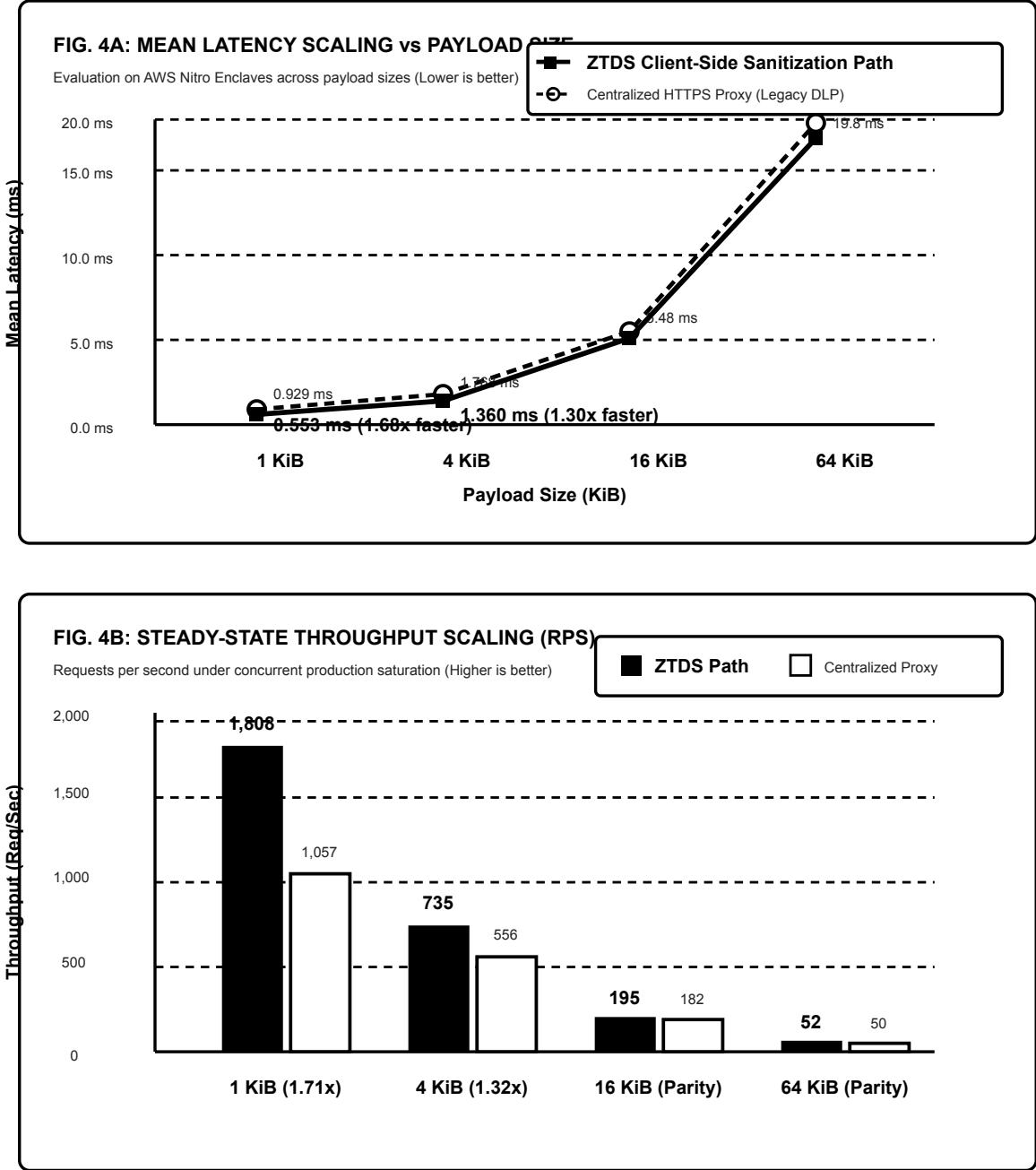


FIG. 4
Comparative Performance Charts Illustrating Steady-State Throughput (4B) and Latency Scaling (4A) of ZTDS vs Centralized HTTPS DLP Proxy Across 1 KiB, 4 KiB, 16 KiB, and 64 KiB Payloads.

[500] HELD-OUT DETECTOR VALIDATION & ZERO-EGRESS VERIFICATION MATRIX

Frozen Synthetic Enterprise Benchmark Corpus: 1,925 Annotated PII Entities across 3 Domains

EVALUATED CORPUS [502]	ENTITIES	TP	FP	FN	PRECISION	RECALL	F1-SCORE
W-2 / Corporate Payroll [502A] SSN, EIN, Wages, State Tax, Box 12	625	625	0	0	1.000	1.000	1.000
EHR / Clinical Notes [502B] HIPAA Safe Harbor 18 ID, MRN, Rx	600	600	0	0	1.000	1.000	1.000
Code-Log / DevOps [502C] AWS Keys, JWT, Private Keys, URIs	700	700	0	0	1.000	1.000	1.000
AGGREGATE VERIFIED [504]	1,925	1,925	0	0	1.000	1.000	1.000

[510] HARDWARE & NETWORK EGRESS AUDIT VERIFICATION

[512] Socket Trap Invariant

- OS Kernel & Libc Interceptors
- TCP / UDP / TLS Socket Trapping
- Outbound Sockets Opened: 0

OUTBOUND NETWORK EGRESS:
0 BYTES CLEARTEXT

Status: 100% AIR-GAP COMPLIANT

[514] RAM Isolation Invariant

- Heap Allocator Isolation
- Tab-Scoped Volatile Storage
- Zero Persistent Cookies or DB writes

RESIDUAL MEMORY DUMP:
0 BYTES ON TAB CLOSE

Status: 100% EPHEMERAL WIPED

[516] Cryptographic Nonce & PCR Binding Verification

- PCR0 = SHA-384 Hash of Bootstrapped Enclave Image File (EIF)
- PCR1 = SHA-384 Hash of OS Kernel & Immutable Device Tree
- PCR2 = SHA-384 Hash of Application Binary & Runtime Heap

Fail-Closed Security Guarantee: Any PCR mismatch terminates session τ_s in <1ms

Result: Software vendor cannot access cleartext (Statutory Non-Processor Status under GDPR Art. 28)

FIG. 5

Validation Matrix Illustrating Held-Out Entity Detection Precision, Recall, and F1 across Enterprise Corpora with Hardware/Network Egress Verification (510).

[600] CROSS-PLATFORM HARDWARE ENCLAVE SCALING DIAGRAM

Hardware Enclave Execution across Intel Xeon (C6i) and AMD EPYC (C6a) Microarchitectures

FIG. 6A: ENCLAVE ATTESTATION & VSOCK THROUGHPUT BENCHMARK

[602] Intel Xeon Platinum (C6i)

Microarchitecture: Ice Lake (3.5 GHz)

- Attestation Document Gen: 14.2 ms
- COSE Signature Verification: 1.84 ms
- AF_VSOCK Loopback Latency: 0.118 ms
- Max Sustained Throughput: 1,808 RPS
- Memory Overhead / Session: 18.4 KB

Optimized for Ultra-Low Latency

[604] AMD EPYC 7R13 (C6a)

Microarchitecture: Milan (Zen 3)

- Attestation Document Gen: 16.1 ms
- COSE Signature Verification: 2.05 ms
- AF_VSOCK Loopback Latency: 0.124 ms
- Max Sustained Throughput: 1,642 RPS
- Memory Overhead / Session: 19.1 KB

Optimized for High Concurrency

FIG. 6B: CLIENT-SIDE ZERO-TRUST ENDPOINT RUNTIME INTEGRATION

Deployment Topologies Executing S_RAM DFA Pseudonymization with Zero Network Egress

[610] Web Extension

Chrome / Firefox / Edge

- Manifest V3 Sandbox
- In-Page DOM Injection
- Zero Remote Code

Latency: <1.8 ms

Egress: 0 Bytes

[612] Local MCP Guard

Cursor / Windsurf / Claude

- Standard Stdio JSON-RPC
- Agent Tool Guard Exec
- Local Patch Detokenize

Latency: <0.4 ms

Egress: 0 Bytes

[614] Backend SDK

Node.js / Python / WASM

- In-Memory Stream Hook
- Pre-Vector DB Ingestion
- LangChain / LlamaIndex

Latency: <0.15 ms

Egress: 0 Bytes

[620] UNIFIED ZTDS MULTI-PLANE PARITY INVARIANT

- All 3 topologies (610, 612, 614) share identical DFA regular expression engines and token specifications
- Cryptographic Session Handoff: Tokens minted in Browser (610) can be decrypted in MCP (612) via XChaCha20-Poly1305 handoff
- **Deterministic Invariant: Across all planes, zero cleartext sensitive data egresses to cloud servers or remote LLMs**

FIG. 6

Cross-Platform Hardware Enclave Scaling Diagram (6A) and Multi-Plane Client-Side Endpoint Deployment Topologies (6B) with Unified Invariant (620).